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# Metric: what does a ONE-SECTOR description retain of the full 240x240 relation
table?
# Inputs: the 240 elements and the {1,11}/{7,13} channel split already used in script
25 (E8-derived, REVIEW input). No 3D/QM content.
import numpy as np, io, contextlib
ns={}
with contextlib.redirect_stdout(io.StringIO()):
exec(open('25_phi_free_pairs.py').read(),ns)
R,P=ns['R'],ns['P']
A=R@np.hstack([P[1],P[11]]); B=R@np.hstack([P[7],P[13]])        # sector coordinates
(4+4)
G=np.round(R@R.T).astype(int); S1=A@A.T; S2=B@B.T
print("check: full relation = sector1 + sector2 products, max residual
%.2e"%np.abs(R@R.T-S1-S2).max())
for name,S in (("{1,11}",S1),("{7,13}",S2)):
    key=np.round(S,6); vals={}
    for k,g in zip(key.ravel(),G.ravel()): vals.setdefault(k,set()).add(g)
    amb=[k for k,v in vals.items() if len(v)>1]
    npair=sum(int((key==k).sum()) for k in amb)
    print(f"sector {name}: distinct sector-product values {len(vals)}; values
compatible with >1 full relation value: {len(amb)}; pairs affected: {npair} of
{G.size} ({100*npair/G.size:.2f}%)")
# linear predictability: can sector-2 coordinates be a real-linear function of
sector-1 coordinates over all 240 elements?
Wl,res,rk,_=np.linalg.lstsq(A,B,rcond=None)
print("best real-linear map sector1->sector2: relative residual %.3f (0 would mean
fully linear-determined); rank of full set %d"%(np.linalg.norm(A@Wl-
B)/np.linalg.norm(B),np.linalg.matrix_rank(R,tol=1e-9)))
```